Balancing Fairness and Utility Constraints for Synthetic Data Generation

Nitish Nagesh
University of California Irvine
nnagesh1@uci.edu

Abstract

Synthetic data generation creates data based on real-world data using generative models. In health applications, generating high-quality data while maintaining fairness for sensitive attributes is essential for equitable outcomes. Existing fairness-aware synthetic data generation models focus on satisfying counterfactual fairness constraints. There is a need to go beyond counterfactual fairness towards causal fairness to retain causal structure during data generation and evaluation. To address this gap, I led the development of two synthetic data generation pipelines - (i) ”FairCauseSyn”, an LLM-based tabular data generator that satisfies causal fairness for health settings (published in IEEE EMBC 2025) and (ii) ”FairTabGen”, a framework that takes into consideration both counterfactual and causal fairness for a wide range of data (submitted to AAAI 2026). My plan is to continue working on enhancing the framework by incorporating effects of unobserved confounders and improving evaluation on metrics such as benefit fairness and process fairness targeting potential submissions in NeurIPS or AAAI in Spring/Summer 2026.

Introduction

Synthetic data has become an essential tool in machine learning, offering a way to mitigate privacy risks, address data scarcity, and promote equity in high-stakes domains such as healthcare, criminal justice, and education (Jordon et al. 2022). Recent advances in generative modeling such as generative adversarial networks (GANs), variational autoencoders (VAEs), diffusion models, and large language models (LLMs) have made it increasingly feasible to generate realistic tabular datasets. However, while models like CTGAN (Xu et al. 2019), TabDDPM (Kotelnikov et al. 2023), REaLTabFormer (Solatorio and Dupriez 2023) and BeGReaT (Borisov et al. 2022) emphasize statistical fidelity, they often fall short in addressing fairness, especially when structural biases are encoded in the training data.

Related Work

Despite advancements in synthetic data generation, fairness remains a critical challenge. While counterfactual fairness has gained popularity as a formal notion of non-discrimination, Schröder, Frauen, and Feuerriegel (2023) argue that this notion is insufficient for practical fairness evaluation due to three structural issues with respect to admissibility, ancestral closure and identifiability. To ameliorate these concerns, another paradigm of causal fairness (Plečko and Bareinboim 2024) has emerged. However, existing fairness-aware synthetic data generation methods such as FairGAN (Xu et al. 2018), DECAF (Van Breugel et al. 2021) primarily focus on satisfying counterfactual fairness constraints (Kusner et al. 2017).

Research Gap

Existing synthetic data generation methods either focus on a single notion of fairness (either counterfactual or causal fairness) and are not geared towards health settings. As illustrated in Fig. 1, there is limited research on exploring examining fairness, bias and utility trade-offs in synthetic data generation from a causal lens for health applications.

Research Question

In my thesis, I aim to answer the following research question:

How can we generate synthetic data that optimizes for causal fairness, counterfactual fairness while retaining data utility and fidelity?

Results as on Sep 30, 2025

I led the development of **FairCauseSyn** (Nagesh, Wang, and Rahmani 2025), an LLM-augmented data generation approach that aims to enhance causal fairness. The experiments on a real-world clinical data shows that the proposed method generates data with less than 10% deviation on fairness metrics. When trained on machine learning predictors, synthetic data reduces bias on sensitive attributes by 70% compared to models trained on real data.

Going beyond causal fairness, I built **FairTabGen** (Nagesh et al. 2025), an LLM-based synthetic data generation framework that incorporates both causal fairness and counterfactual fairness in the data generation and evaluation phase. Across datasets in the law school admissions, COMPAS and MIMIC, the method outperforms state-of-the-art by 10% on fairness-related metrics while retaining statistical utility

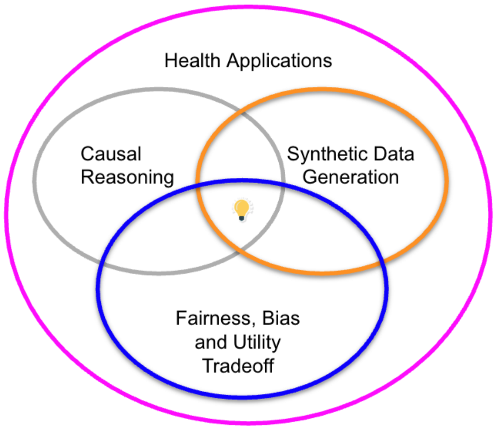

Figure 1: Research Themes and Intersectionality

while using less than 20% of original data, demonstrating efficiency in low-data regimes.

Planned contributions by Jan 20, 2026

I am working towards revising my submission about balancing counterfactual fairness and causal fairness along with utility constraints from AAAI. The components to add would be more related baselines, enhance the synthetic data generation algorithm and perform extensive experiments for health-related case studies. I plan to submit the revised work to the International Conference on Machine Learning (ICML) by January 30, 2026.

Targeted submission by Aug 15, 2026

In the future, I aim to extend this work further into two strands. The first one would be to simulate the effect of unobserved confounders using the extended fairness map into the data generation process (Plečko and Bareinboim 2024; Schröder, Frauen, and Feuerriegel 2023). Another strand is to enhance the evaluation to analyze fairness from a decision maker perspective via benefit fairness (Plečko and Bareinboim 2024) to ensure synthetically generated data is beneficial for advancing health equity. I plan to submit this work either to NeurIPS in May 2026 or AAAI in August 2026 depending on my results and progress by that time.

Resources and Mentorship

I will continue working with my advisor Prof. Amir Rahmani from University of California Irvine, who has expertise in multimodal conversational health agents, and my collaborator Dr. Randall Stafford from Stanford, who provides clinical expertise on fairness and bias in healthcare. I have applied for the NVIDIA fellowship and internal department-wide fellowships. I will leverage computational capabilities to run large scale models to generate data for electronic health record (EHR)-style datasets such as MIMIC-IV and proprietary observational health data to validate my approach.

Anticipated Thesis Contribution

The goal of my thesis to develop and evaluate synthetic data generation techniques that take into consideration fairness and utility metrics both in the data generation and evaluation process. Doing so would allow the generated data to be used for downstream machine learning prediction tasks with limited bias. Since the application of my framework is mostly in health settings and generative models are on the rise, being cognizant of fairness in the entire machine learning pipeline will contribute to improving health equity.

References

Borisov, V.; Seßler, K.; Leemann, T.; Pawelczyk, M.; and Kasneci, G. 2022. Language models are realistic tabular data generators. *arXiv preprint arXiv:2210.06280*.

Jordon, J.; Szpruch, L.; Houssiau, F.; Bottarelli, M.; Cherubin, G.; Maple, C.; Cohen, S. N.; and Weller, A. 2022. Synthetic Data–what, why and how? *arXiv preprint arXiv:2205.03257*.

Kotelnikov, A.; Baranchuk, D.; Rubachev, I.; and Babenko, A. 2023. Tabddpm: Modelling tabular data with diffusion models. In *International conference on machine learning*, 17564–17579. PMLR.

Kusner, M. J.; Loftus, J.; Russell, C.; and Silva, R. 2017. Counterfactual fairness. *Advances in neural information processing systems*, 30.

Nagesh, N.; Shakibhamedan, S.; Bagheri, M.; Wang, Z.; TaheriNejad, N.; Jantsch, A.; and Rahmani, A. M. 2025. FairTabGen: Unifying Counterfactual and Causal Fairness in Synthetic Tabular Data Generation. *arXiv preprint arXiv:2508.11810*.

Nagesh, N.; Wang, Z.; and Rahmani, A. M. 2025. FairCauseSyn: Towards Causally Fair LLM-Augmented Synthetic Data Generation. *arXiv preprint arXiv:2506.19082*.

Plečko, D.; and Bareinboim, E. 2024. Causal Fairness Analysis: A Causal Toolkit for Fair Machine Learning. *Foundations and Trends® in Machine Learning*, 17(3): 304–589.

Schröder, M.; Frauen, D.; and Feuerriegel, S. 2023. Causal fairness under unobserved confounding: A neural sensitivity framework. *arXiv preprint arXiv:2311.18460*.

Solatorio, A. V.; and Dupriez, O. 2023. Realtabformer: Generating realistic relational and tabular data using transformers. *arXiv preprint arXiv:2302.02041*.

Van Breugel, B.; Kyono, T.; Berrevoets, J.; and Van der Schaar, M. 2021. Decaf: Generating fair synthetic data using causally-aware generative networks. *Advances in Neural Information Processing Systems*, 34: 22221–22233.

Xu, D.; Yuan, S.; Zhang, L.; and Wu, X. 2018. Fairgan: Fairness-aware generative adversarial networks. In *2018 IEEE international conference on big data (big data)*, 570–575. IEEE.

Xu, L.; Skoularidou, M.; Cuesta-Infante, A.; and Veeramachaneni, K. 2019. Modeling tabular data using conditional gan. *Advances in neural information processing systems*, 32.